

COPPER TAPE LIGHTHOUSE

Light Up Your Project!

THE STORY OF THE SEA

WHY DO SHIPS NEED LIGHTS?

Imagine standing on a beach at night. The sea is dark and dangerous! Ships out in the ocean need a way to find their way safely back to the shore.

LIGHTHOUSES: THE TALL GUIDES

Lighthouses are tall towers that shine a bright beam of light across the water. In India, Coastal Guards use these towers to help guide boats through the darkness.

Today, we are going to build our own lighthouse using a "Path of Electricity"!

Let's turn on the lights!

Meet the Tools

To make electricity flow, we need a few "Circuit Friends." Let's learn their names:

LED – The "Small Light"

This is a tiny lamp. It is very special because it only lights up when electricity flows through it the right way!

Coin Cell Battery – The "Power Button"

This little silver disc holds all the energy. It is the heart of our circuit!

Copper Tape – The "Shiny Tracks"

Electricity loves to travel on copper. We will use this sticky tape to build a road for the energy to follow.

What is a Circuit?

Electricity flows just like water in a pipe! A circuit is a Path that must be "Bandh" (Closed) for electricity to move.

Think of it as a Loop:

- It starts at the Battery.
- Travels through the Shiny Copper Tape.
- Lights up the LED.
- Goes all the way back to the Battery!

If the loop is broken, the light goes out!

INDIAN LIGHTHOUSES

Amazing Coastal Towers

India has many famous lighthouses that protect our coastline. Have you heard of these?

Dolphin's Nose

Located in Visakhapatnam, this lighthouse sits high on a hill that looks like a dolphin's nose!

Aguada Lighthouse

A very old and famous lighthouse in Goa that has watched over ships for many years.

Did you know? Many lighthouses in Gujarat and Kerala are painted with bold Red-and-White stripes!

Building the Track

It's time to become an Electricity Detective!

Step 1: Peel and Stick

Stick your copper tape along the dotted lines on your lighthouse template.

Step 2: Press Firmly

Use your fingers to press down hard. Make sure your "Shiny Track" is continuous with no gaps or breaks!

Step 3: Connect the Friends

Place your coin battery on the tape (Positive + side up!). Put the LED legs on the tracks: Long leg on Positive (+), Short leg on Negative (-).

CHAMATKAR! (MAGIC)

Press down gently and watch what happens...

Detective Challenge!

Light not glowing? Check your path:

- Are there any gaps in the copper tape?
- Is the battery flat side up?
- Try flipping the LED around!

Success Criteria:

- ✓ I can identify the battery, tape, and LED.
- ✓ I know electricity needs a complete loop.
- ✓ My lighthouse LED glows bright!